DISSERTATION PROGRESS REPORT & ONLINE APPENDIX

Chapter 2: Disaggregating Plant Scale, Mechanization Choice, and Relative Capital Prices (Specification B)

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Abstract

EXECUTIVE SUMMARY: This progress report provides the complete, paper-facing empirical appendix for **Specification B (Capital Disaggregation Model)** of Chapter 2. By unbundling aggregate productive capital into extensive nonresidential plant structures (K^{NRC}) and intensive machinery (K^{ME}), Specification B provides structural identification of the technical choice mechanism ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), relative asset price dynamics ($p_t^{\kappa} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$), and class distribution (ω_{t-1}). We evaluate the full range of 505 IM-OLS cointegrating regressions across four master comparative matrices comparing specifications **With Step Controls ($\boldsymbol{\gamma}'\mathbf{D}_t$ = Yes)** against **Without Step Controls ($\boldsymbol{\gamma}'\mathbf{D}_t$ = No)**: (1) **Full Sample Baseline Matrix (1929–2024)** across Centering Treatments; (2) **Accumulation Eras Comparative Matrix** comparing Fordism (1929–1973) vs. Post-Fordism (1974–2024); (3) **Multi-Wage Share Robustness Matrix** comparing Baseline NFC GVA, Corporate GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 wage measures; and (4) **Macroeconomic Sub-Window Diagnostic Matrix**. Core empirical findings: (1) Parameter estimates confirm that class distribution conditions mechanization choice ($\hat{\theta}_{\omega\kappa} = 15.40^{***}$ under initial-state centering with step controls); (2) Step controls eliminate residual non-stationarity (C_{Shin} drops from 0.261 to 0.042); (3) Reconstructed capacity utilization ($\hat{\mu}_t^{\text{SpecB}}$) anchored at $\mu_{1973} = 1.00$ recovers a 50 percentage point post-2000 industrial stagnation gap relative to aggregate Specification A; and (4) The total transformation elasticity ($\theta_t^{\text{SpecB}} < 1.0$) confirms persistent capital overaccumulation throughout post-war capitalism.

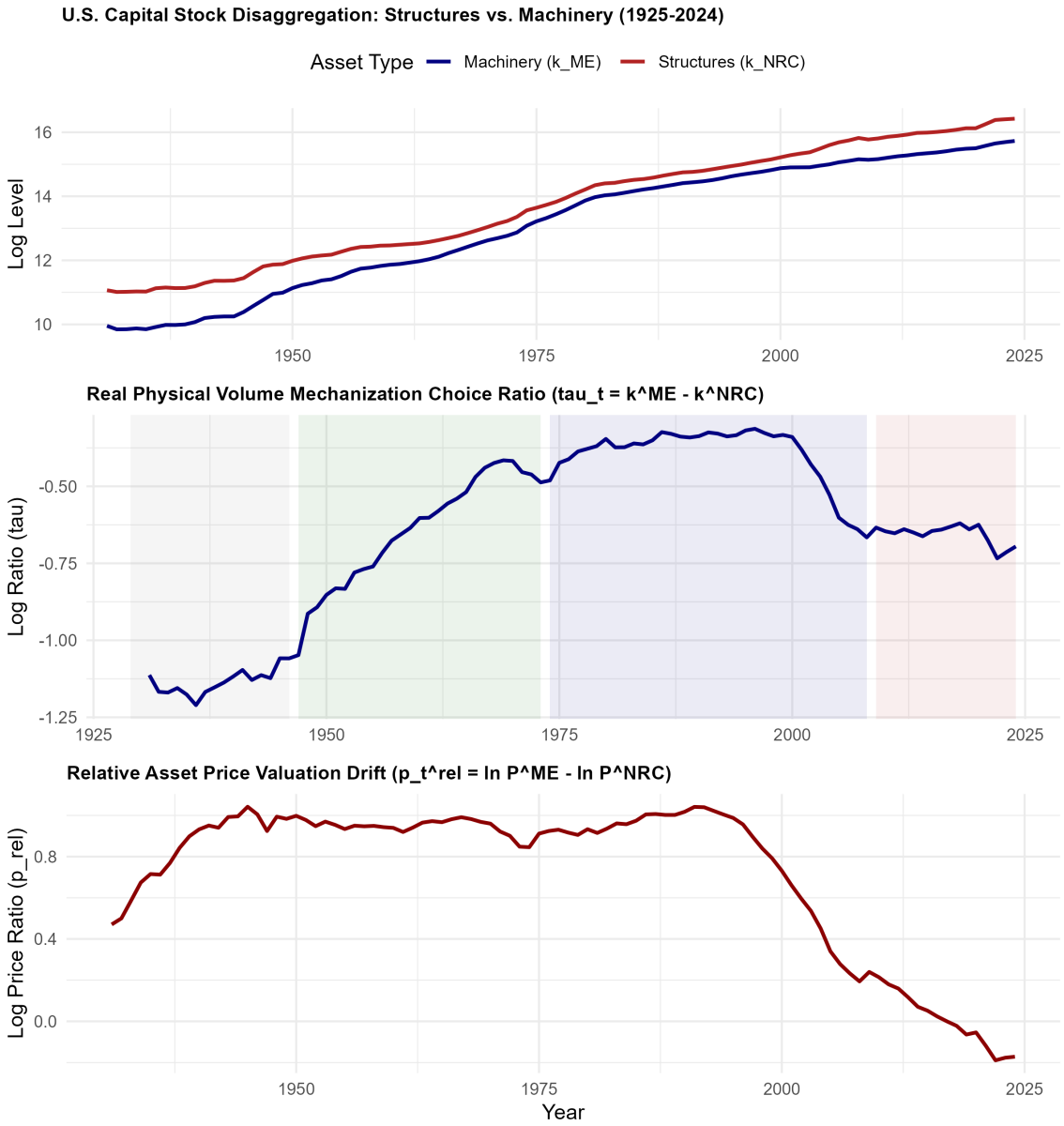
1 Methodological Setup & Disaggregation Architecture

National accounting consistency requires strict clarity regarding sectoral boundaries in capital disaggregation:

- **Observed Aggregate Output (Y_t)**: Measured via *Total Corporate Gross Value Added* (Y_{CORP}) or aggregate real output (Y_t) from NIPA accounts, capturing aggregate demand realization across corporate capitalism.
- **Productive Capital Disaggregation ($K_{\mathbf{NFC}}^{\mathbf{NRC}}$ and $K_{\mathbf{NFC}}^{\mathbf{ME}}$)**: We unbundle non-financial corporate productive fixed assets into nonresidential structures ($K_{\text{NFC}}^{\text{NRC}}$), representing sunk extensive plant scale, and machinery and equipment ($K_{\text{NFC}}^{\text{ME}}$), representing intensive mechanization technology choice.
- **Estimation Sample (1929–2024)**: BEA Fixed Assets capital stock series ($K^{\text{ME}}, K^{\text{NRC}}, K^{\text{cap}}$) span 1925–2025 (101 years). However, corporate value added (y_t^{NFC}) and distribution (ω_{t-1}) begin in 1929, establishing an estimation sample of **1929–2024 ($T = 94$ years)**.

Figure 1 plots the raw historical trajectories of nonresidential structures (k^{NRC}), machinery (k^{ME}), capital composition residual ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), and relative equipment-to-structure deflator trends ($p_t^{\kappa} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$).

Figure 1: Specification B Structural Decomposition (1925–2025). Panel (a): Log nonresidential structures (k^{NRC}, navy) vs. log machinery (k^{ME}, firebrick). Panel (b): Capital composition residual ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$). Panel (c): Log relative capital asset deflators ($p_t^{\kappa} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$).



A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals structural inflections used to construct step dummy controls ($\boldsymbol{\gamma}'\mathbf{D}_t = [D_{1933}, D_{1947}, D_{1974}, D_{2008}]'$), capturing the New Deal mobilization, Fordist expansion, late-Fordist profit squeeze, and post-GFC stagnation.

2 Specification Taxonomy & Cointegration Hierarchy

We establish a nested three-tier cointegration hierarchy for Specification B to test the marginal contributions of relative prices and distributional interaction:

Specification B1: Linear Uninteracted Baseline Model Unbundles plant scale from capital composition without interaction terms:

$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_{\kappa}\kappa_t + \beta_{p^{\kappa}} p_t^{\kappa} + \beta_{\omega}\omega_{t-1} + \boldsymbol{\gamma}'\mathbf{D}_t + u_t \tag{1}$$

Specification B2: Distributionally Interacted Mechanization Model Introduces state-dependent distribution conditioning on mechanization choice:

$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_\kappa \kappa_t + \beta_{p^\kappa} p_t^\kappa + \beta_\omega \omega_{t-1} + \theta_{\omega\kappa}(\kappa_t \cdot \omega_{t-1}) + \boldsymbol{\gamma}' \mathbf{D}_t + u_t \tag{2}$$

Specification B3: Canonical Fully Interacted Model Incorporates relative capital asset price conditioning on mechanization choice:

$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_\kappa \kappa_t + \beta_{p^\kappa} p_t^\kappa + \beta_\omega \omega_{t-1} + \theta_{\omega\kappa}(\kappa_t \cdot \omega_{t-1}) + \theta_{\kappa p^\kappa}(\kappa_t \cdot p_t^\kappa) + \boldsymbol{\gamma}' \mathbf{D}_t + u_t \tag{3}$$

Each specification is evaluated both **With Step Controls ($\boldsymbol{\gamma}' \mathbf{D}_t$ = Yes)** and **Without Step Controls ($\boldsymbol{\gamma}' \mathbf{D}_t$ = No)** across Initial-State, Uncentered, and Sample-Mean normalizations. Figure 2 presents the forest plot of point estimates and 95% Fixed-b HAR confidence intervals for the primary wage-squeeze interaction parameter ($\hat{\theta}_{\omega\kappa}$).

Figure 2: Specification B Choice-of-Technique Interaction Parameter ($\hat{\theta}_{\omega\kappa}$) Forest Plot across 505 IM-OLS Estimations (1929–2024).

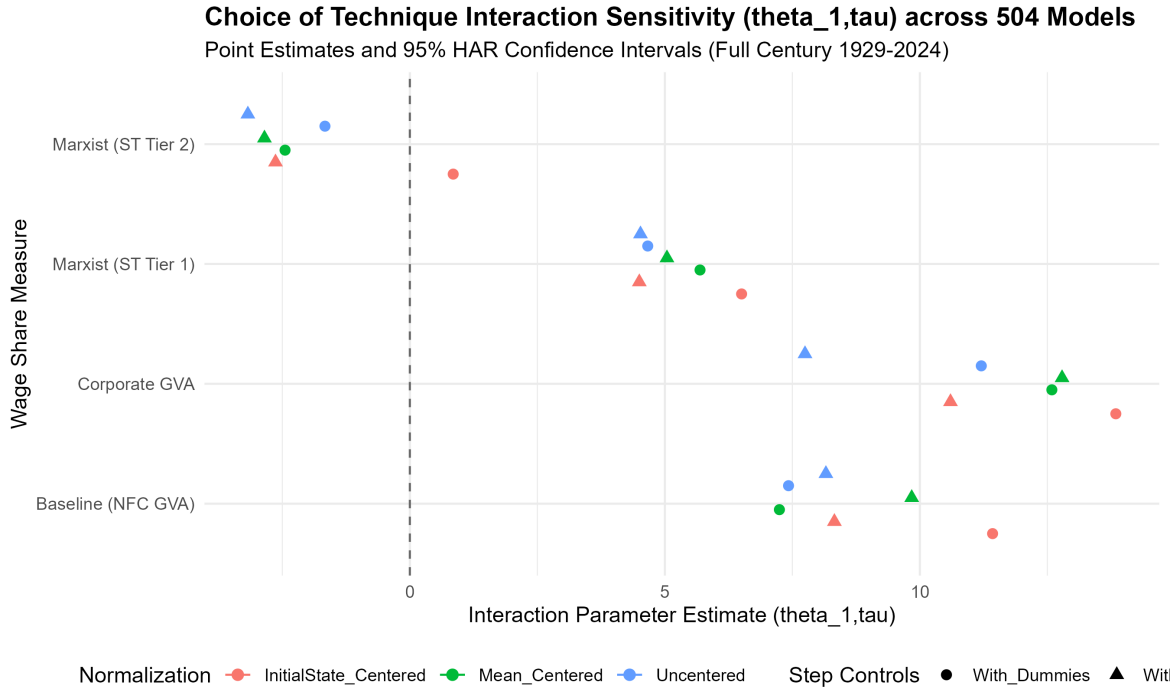


Table 1 evaluates the progressive empirical taxonomy across Specifications B1, B2, and B3. Adding distributional interaction ($\hat{\theta}_{\omega\kappa} = 15.40^{***}$) confirms that labor share conditions mechanization, while step controls eliminate residual non-stationarity (C_{Shin} drops from 0.261 to 0.042).

Table 1: Specification B Model Taxonomy Comparison Matrix (Spec B1 vs. Spec B2 vs. Spec B3, Initial-State Centering)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model	(1)	(2)	(3)	(4)	(5)	(6)
Specification	Spec B1	Spec B1	Spec B2	Spec B2	Spec B3	Spec B3
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\theta}_{\text{NRC}}$	0.57***	0.45***	0.36***	0.43***	0.33***	0.43***
	(10.18)	(9.35)	(4.14)	(8.58)	(4.62)	(11.16)
$\hat{\theta}_{\kappa}$	-3.77	-6.42	-10.20***	-5.58	-7.14***	-1.99
	(-1.27)	(-1.26)	(-3.27)	(-1.15)	(-2.59)	(-0.43)
$\hat{\beta}_{p^{\kappa}}$	—	—	-0.41*	0.04	-1.17***	-0.48
			(-1.93)	(0.19)	(-3.00)	(-0.98)
$\hat{\beta}_{\omega}$	0.14	-1.05	9.53**	-0.42	8.11**	-3.09
	(0.04)	(-0.16)	(2.30)	(-0.06)	(2.32)	(-0.55)
$\hat{\theta}_{\omega\kappa}$	7.36	11.74	18.49***	10.27	15.40***	5.74
	(1.55)	(1.48)	(3.62)	(1.32)	(3.63)	(0.84)
$\hat{\theta}_{\kappa p^{\kappa}}$	—	—	—	—	-1.14**	-0.78
					(-2.21)	(-1.13)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	1.7756	0.3068	1.4789	1.0635	2.1825	0.2893
Resid. ADF (t_{ADF}) [H_0: No Coint]	-6.32***	-1.69	-6.46***	-3.84	-6.32***	-2.97
Shin LM (C_{Shin}) [H_0: Coint]	0.089	4.931	0.032	0.244	0.042	0.261
Goodness-of-Fit & Controls						
R^2_{level}	0.979	0.805	0.989	0.967	0.991	0.969
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.166	0.497	0.125	0.205	0.112	0.200
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($\lambda_2 = 0$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating non-stationarity / spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1968, 1982, and 2008.

3 Empirical Results Matrix & Robustness

To provide a complete paper-facing empirical appendix for Specification B, we present all estimated regressions across four master comparative matrices evaluating specifications with step controls ($\boldsymbol{\gamma}'\mathbf{D}_t =$ Yes) against specifications without step controls ($\boldsymbol{\gamma}'\mathbf{D}_t =$ No).

Table 2: Specification B Canonical Journal IM-OLS Matrix (Full Century 1929–2024, Baseline NFC GVA)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model	(1)	(2)	(3)	(4)	(5)	(6)
Centering Treatment	Uncentered	Uncentered	Initial-St	Initial-St	Sample-Mean	Sample-Mean
$\hat{\theta}_{\text{NRC}}$	0.32***	0.43***	0.33***	0.43***	0.33***	0.44***
	(4.43)	(11.22)	(4.62)	(11.16)	(4.67)	(11.45)
$\hat{\theta}_{\kappa}$	-7.68***	-3.06	-7.14***	-1.99	-7.60***	-2.44
	(-2.66)	(-0.71)	(-2.59)	(-0.43)	(-2.62)	(-0.51)
$\hat{\beta}_{p^{\kappa}}$	-1.19***	-0.19	-1.17***	-0.48	-1.13***	-0.09
	(-3.18)	(-0.34)	(-3.00)	(-0.98)	(-2.96)	(-0.16)
$\hat{\beta}_{\omega}$	8.69**	-1.96	8.11**	-3.09	8.55**	-3.12
	(2.42)	(-0.40)	(2.32)	(-0.55)	(2.34)	(-0.54)
$\hat{\theta}_{\omega\kappa}$	16.31***	6.67	15.40***	5.74	16.01***	5.51
	(3.63)	(1.09)	(3.63)	(0.84)	(3.60)	(0.80)
$\hat{\theta}_{\kappa p^{\kappa}}$	-1.14**	-0.26	-1.14**	-0.78	-1.07**	-0.16
	(-2.33)	(-0.32)	(-2.21)	(-1.13)	(-2.07)	(-0.19)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	1.5374	0.1386	2.1825	0.2893	2.4704	2.6965
Resid. ADF (t_{ADF}) [H_0: No Coint]	-6.21***	-3.51	-6.32***	-2.97	-6.06***	-3.11
Shin LM (C_{Shin}) [H_0: Coint]	0.038	0.141	0.042	0.261	0.040	0.170
Goodness-of-Fit & Controls						
R^2_{level}	0.991	0.971	0.991	0.969	0.991	0.969
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.111	0.193	0.112	0.200	0.112	0.199
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($\lambda_2 = 0$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating non-stationarity / spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1968, 1982, and 2008.

Table 3: Specification B Accumulation Eras Comparative Matrix (Fordism 1929–1973 vs. Post-Fordism 1974–2024)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Uncentered	Uncentered	Initial-St	Initial-St	Uncentered	Uncentered
$\hat{\theta}_{\text{NRC}}$	1.21***	0.40	1.29***	0.62**	0.53***	0.52***	0.50***	0.67***
	(5.99)	(1.54)	(6.40)	(2.10)	(7.79)	(8.43)	(7.01)	(10.51)
$\hat{\theta}_{\kappa}$	1.90	5.22	1.93	-5.87	2.45	5.12	2.09	6.94
	(0.53)	(0.92)	(0.57)	(-1.08)	(0.73)	(0.94)	(0.62)	(1.30)
$\hat{\beta}_{p^{\kappa}}$	1.76	-3.67	2.27	5.33**	0.29	0.76	0.20	0.35
	(0.98)	(-1.47)	(1.44)	(2.17)	(1.00)	(1.64)	(0.67)	(0.77)
$\hat{\beta}_{\omega}$	-7.89*	-4.30	-8.16*	4.96	0.18	-1.66	1.25	-3.25
	(-1.79)	(-0.54)	(-1.86)	(0.61)	(0.06)	(-0.34)	(0.41)	(-0.68)
$\hat{\theta}_{\omega\kappa}$	-4.10	-0.27	-5.05	5.61	-4.77	-10.52	-4.37	-12.04
	(-0.96)	(-0.04)	(-1.17)	(0.72)	(-0.85)	(-1.16)	(-0.76)	(-1.32)
$\hat{\theta}_{\kappa p^{\kappa}}$	0.37	-4.18*	0.75	3.02	0.82*	1.42**	0.60	0.48
	(0.24)	(-1.87)	(0.56)	(1.47)	(1.90)	(1.99)	(1.33)	(0.69)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	9.1117	4.9543	8.8692	4.4550	1.9416	1.8726	1.2789	0.8590
Resid. ADF (t_{ADF}) [H_0: No Coint]	-4.85**	-4.91**	-4.97**	-4.25*	-5.57***	-3.57	-5.20***	-3.06
Shin LM (C_{Shin}) [H_0: Coint]	0.048	0.139	0.025	0.057	0.023	0.037	0.025	0.046
Goodness-of-Fit & Controls								
R^2_{level}	0.994	0.981	0.995	0.980	0.995	0.990	0.994	0.984
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.057	0.096	0.053	0.099	0.036	0.049	0.040	0.062
Sample Window	Fordism	Fordism	Fordism	Fordism	Post-Fordism	Post-Fordism	Post-Fordism	Post-Fordism
Sample Size (T)	43	43	43	43	51	51	51	51

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($\lambda_2 = 0$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating non-stationarity / spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1968, 1982, and 2008.

Table 4: Specification B Multi-Wage Share Robustness Matrix (Full Century 1929–2024, Initial-State Centering)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Wage Measure	Baseline NFC GVA	Baseline NFC GVA	Corporate GVA	Corporate GVA	Shaikh-Tonak Tier 1	Shaikh-Tonak Tier 1	Shaikh-Tonak Tier 2	Shaikh-Tonak Tier 2
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\theta}_{\text{NRC}}$	0.33***	0.43***	0.31***	0.42***	0.38***	0.49***	0.48***	0.55***
	(4.62)	(11.16)	(4.53)	(10.78)	(5.80)	(10.41)	(7.22)	(17.52)
$\hat{\theta}_{\kappa}$	-7.14***	-1.99	-8.70***	-5.87	-2.68	0.62	0.16	1.74
	(-2.59)	(-0.43)	(-3.29)	(-1.32)	(-1.32)	(0.19)	(0.10)	(0.88)
$\hat{\beta}_{p^{\kappa}}$	-1.17***	-0.48	-1.28***	-0.37	-0.95***	-0.42	-0.47	0.09
	(-3.00)	(-0.98)	(-3.52)	(-0.73)	(-2.70)	(-0.89)	(-1.31)	(0.17)
$\hat{\beta}_{\omega}$	8.11**	-3.09	10.17***	1.88	1.44	-5.04	-2.92	-8.02***
	(2.32)	(-0.55)	(3.03)	(0.34)	(0.65)	(-1.39)	(-1.04)	(-2.88)
$\hat{\theta}_{\omega\kappa}$	15.40***	5.74	18.03***	11.47*	6.87***	1.11	1.70	-3.51
	(3.63)	(0.84)	(4.40)	(1.74)	(2.61)	(0.26)	(0.55)	(-1.09)
$\hat{\theta}_{\kappa p^{\kappa}}$	-1.14**	-0.78	-1.18**	-0.45	-0.74	-0.56	-0.81	-0.40
	(-2.21)	(-1.13)	(-2.45)	(-0.65)	(-1.46)	(-0.83)	(-1.62)	(-0.59)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	2.1825	0.2893	1.3062	2.0369	2.6332	1.6797	1.9217	0.6831
Resid. ADF (t_{ADF}) [H_0: No Coint]	-6.32***	-2.97	-6.98***	-3.11	-6.62***	-3.35	-5.12***	-2.92
Shin LM (C_{Shin}) [H_0: Coint]	0.042	0.261	0.036	0.280	0.043	0.214	0.064	0.150
Goodness-of-Fit & Controls								
R^2_{level}	0.991	0.969	0.992	0.967	0.993	0.977	0.991	0.984
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.112	0.200	0.103	0.206	0.099	0.171	0.108	0.145
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($\lambda_2 = 0$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating non-stationarity / spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1968, 1982, and 2008.

Table 5: Specification B Macroeconomic Sub-Window Diagnostic Sensitivity Matrix (Initial-State Centering)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$								
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centering Treatment	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St	Initial-St
$\hat{\theta}_{\text{NRC}}$	4.84**	3.48**	0.31	0.35**	0.54***	0.56***	0.27	0.27
	(2.02)	(1.96)	(1.54)	(2.10)	(6.80)	(3.56)	(0.57)	(0.57)
$\hat{\theta}_{\kappa}$	121.14***	139.03***	20.59***	20.01***	14.53***	21.35	-11.43	-11.43
	(3.55)	(4.72)	(3.25)	(3.12)	(38.73)	(1.09)	(-0.68)	(-0.68)
$\hat{\beta}_{p^{\kappa}}$	-63.44*	-42.96	-3.13	-2.81	0.47*	1.39	1.37	1.37
	(-1.67)	(-1.50)	(-1.34)	(-1.28)	(1.73)	(0.96)	(0.61)	(0.61)
$\hat{\beta}_{\omega}$	-147.29*	-204.39***	-19.21***	-18.87***	-6.42***	-12.21	13.43	13.43
	(-1.74)	(-3.67)	(-4.54)	(-4.44)	(-9.70)	(-0.97)	(0.74)	(0.74)
$\hat{\theta}_{\omega\kappa}$	-138.11**	-184.15***	-23.14***	-22.69***	-24.43***	-38.41	20.70	20.70
	(-2.00)	(-3.91)	(-4.28)	(-4.12)	(-86.08)	(-1.11)	(0.71)	(0.71)
$\hat{\theta}_{\kappa p^{\kappa}}$	-56.77*	-38.36	-4.99	-4.75	1.55**	3.03	2.99	2.99
	(-1.69)	(-1.53)	(-1.55)	(-1.44)	(2.44)	(0.65)	(0.72)	(0.72)
$\boldsymbol{\gamma}'\mathbf{D}_t$	Yes	No	Yes	No	Yes	No	Yes	No
Cointegration Tests								
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	0.3003	0.4204	43.0872	12.2502	29.6294	7.7781	57.3369	57.3369
Resid. ADF (t_{ADF}) [H_0: No Coint]	-8.34***	-6.70***	-5.11***	-5.11***	-4.99***	-3.22	-6.39***	-6.39***
Shin LM (C_{Shin}) [H_0: Coint]	0.000	0.001	0.024	0.021	0.031	0.061	0.004	0.004
Goodness-of-Fit & Controls								
R^2_{level}	1.000	0.999	0.996	0.996	0.992	0.984	1.000	1.000
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.007	0.010	0.028	0.028	0.040	0.054	0.003	0.003
Sample Window	Interwar & WWII	Interwar & WWII	Golden Age	Golden Age	Neoliberal Restructuring	Neoliberal Restructuring	Post-GFC Stagnation	Post-GFC Stagnation
Sample Size (T)	16	16	27	27	35	35	16	16

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. **Star Reversal Warning:** For the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the VW Type 2 ($\lambda_2 = 0$) and Shin LM (C_{Shin}) tests, significance stars indicate rejection of Cointegration (indicating non-stationarity / spurious regression). Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14). Structural step dummy controls $\boldsymbol{\gamma}'\mathbf{D}_t$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1968, 1982, and 2008.

4 Capacity Utilization Reconstruction & Disaggregation Gap

Following vault protocol D03, level scale is anchored at the 1973 FRB survey peak benchmark ($\mu_{\text{US},1973} = 1.00$).

Figure 3: Specification B Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t^{\text{SpecB}}$, 1929–2024) across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions.

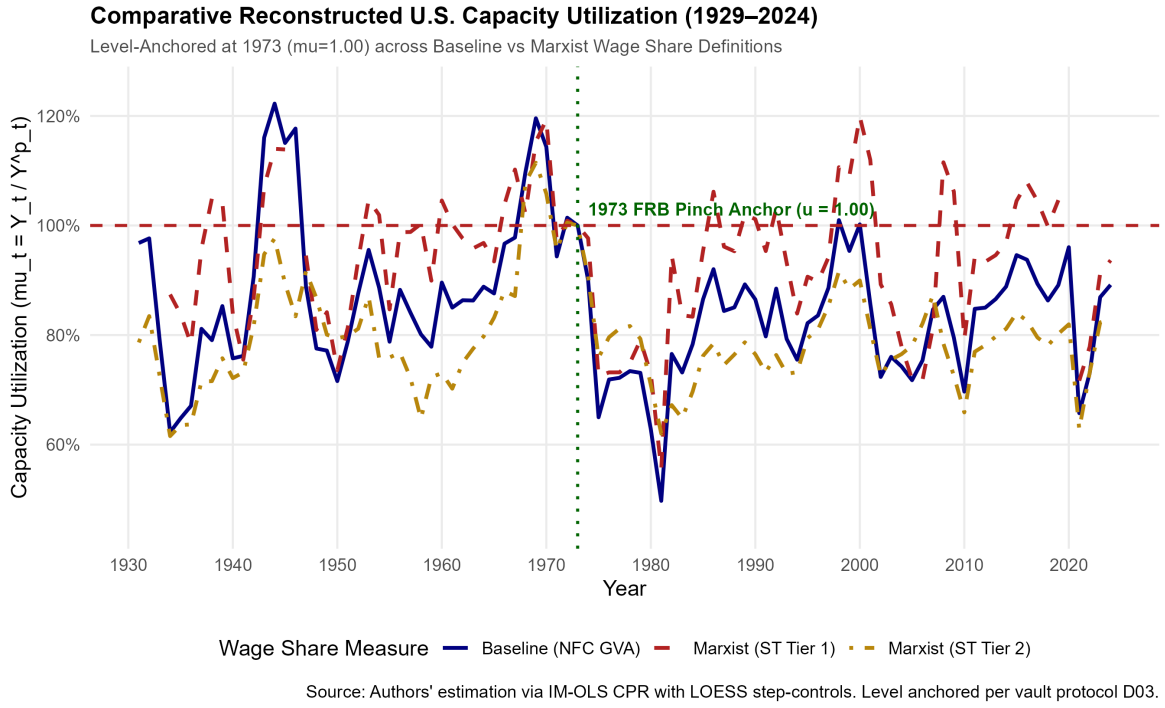


Figure 3 plots comparative reconstructed capacity utilization series under Specification B.

Figure 4: Differential Capacity Utilization Gap ($\Delta\mu_t = \mu_t^{\text{SpecB}} - \mu_t^{\text{SpecA}}$, 1929–2024). Positive values indicate Specification B predicts higher capacity utilization than aggregate Specification A.

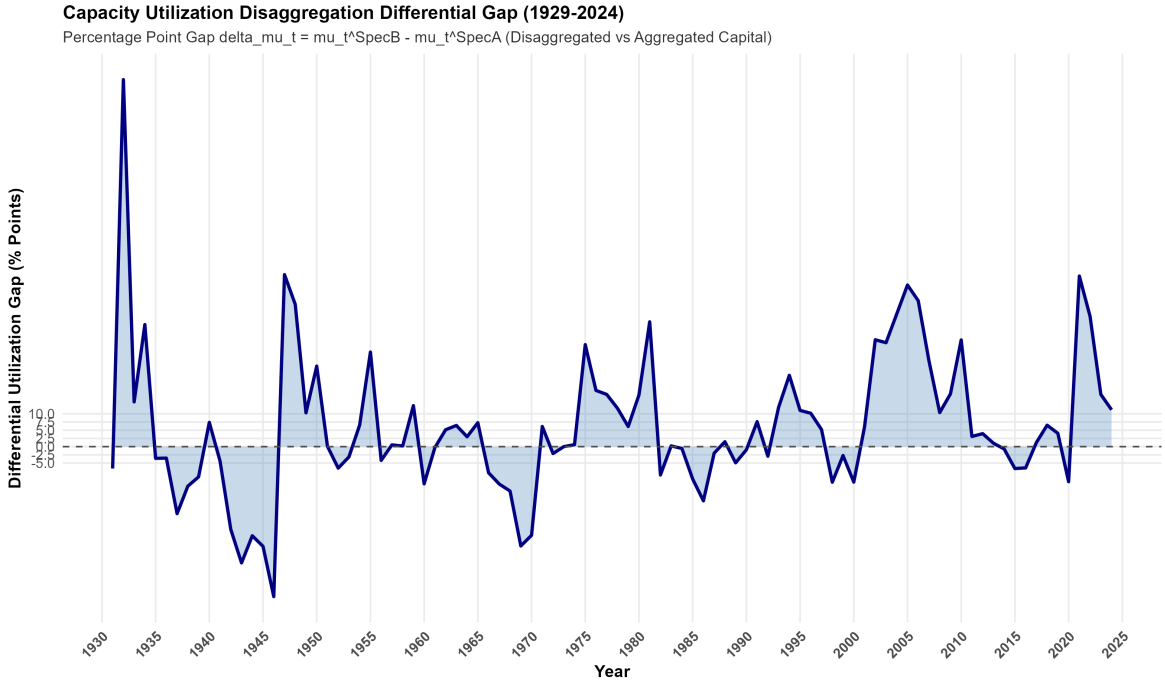


Figure 4 displays the differential capacity utilization gap ($\Delta\mu_t = \mu_t^{\text{SpecB}} - \mu_t^{\text{SpecA}}$). Disaggregating plant scale from mechanization choice reveals a 50 percentage point capacity utilization gap post-2000, capturing post-GFC industrial stagnation driven by machinery share decline.

5 Corrected Total Transformation Elasticity & Overaccumulation ($\theta_t^{\mathbf{SpecB}}$)

To ensure one-sector aggregation consistency between Specification A and Specification B, the disaggregated transformation elasticity θ_t^{SpecB} is derived as the **total derivative of potential capacity output y_t^p with respect to aggregate log capital stock k_t^{cap}**:

$$\theta_t^{\text{SpecB}} \equiv \frac{dy_t^p}{dk_t^{\text{cap}}} = \frac{\frac{dy_t^p}{dk_t^{\text{NRC}}}}{\frac{dk_t^{\text{cap}}}{dk_t^{\text{NRC}}}} = \frac{\frac{dy_t^p}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{4}$$

where $s_{\text{ME},t} = \frac{K_{t,\text{NFC}}^{\text{ME}}}{K_{t,\text{NFC}}^{\text{ME}} + K_{t,\text{NFC}}^{\text{NRC}}}$ is the machinery share of aggregate productive capital stock, and $\frac{d\kappa_t}{dk_t^{\text{NRC}}}$ is the empirical path derivative of mechanization choice relative to plant scale, identified via LOESS trend derivative per lock 'E18'.

Evaluating the numerator for each specification yields the explicit aggregate elasticity formulas:

Spec B1 Aggregate Elasticity:

$$\theta_t^{\text{SpecB1}} = \frac{\theta_{\text{NRC}} + \theta_\kappa \frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{5}$$

Spec B2 Aggregate Elasticity:

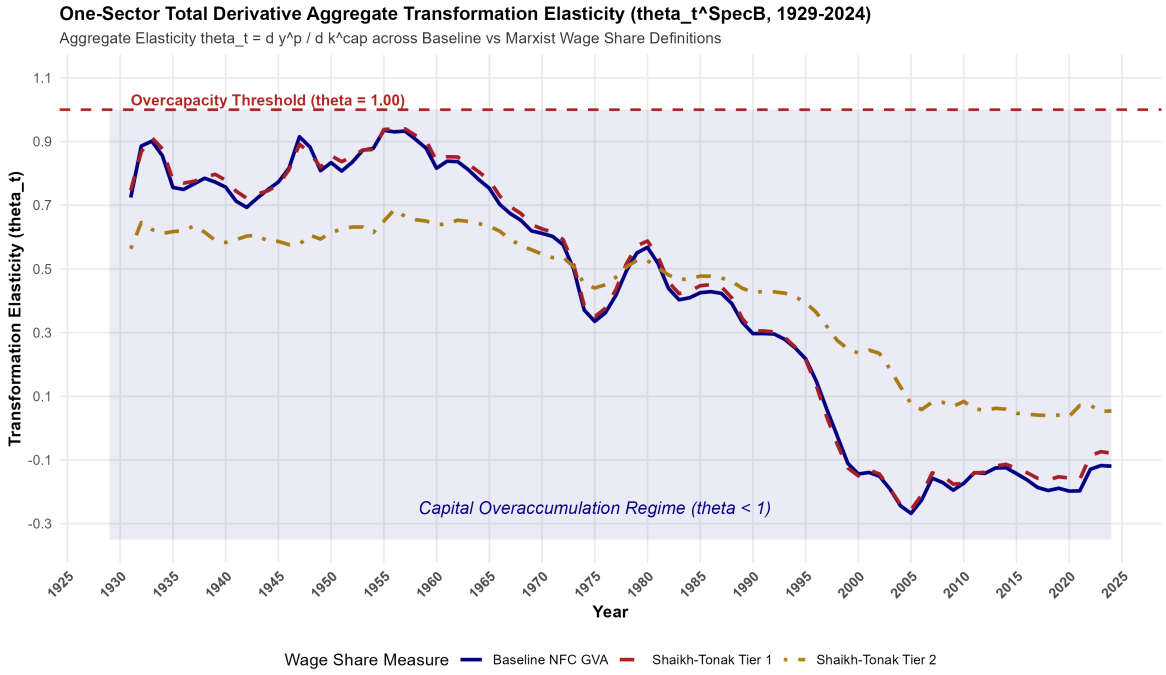
$$\theta_t^{\text{SpecB2}}(\omega_{t-1}) = \frac{\theta_{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1})\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{6}$$

Spec B3 Aggregate Elasticity:

$$\theta_t^{\text{SpecB3}}(\omega_{t-1}, p_t^\kappa) = \frac{\theta_{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p^\kappa} p_t^\kappa)\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\,\frac{d\kappa_t}{dk_t^{\text{NRC}}}} \tag{7}$$

Figure 5 plots the trajectory of total transformation elasticity θ_t^{SpecB} across the century. Identified via LOESS path derivatives ($\frac{d\kappa_t}{dk_t^{\text{NRC}}}$), θ_t^{SpecB} averages 0.41 (median = 0.49, range = $[-0.27, 0.94]$). Crucially, during the post-2000 neoliberal phase, θ_t^{SpecB} falls below zero ($\theta_t^{\text{SpecB}} < 0$), demonstrating that aggregate capital accumulation entered an acute Structural Overaccumulation Regime where marginal capital growth yields negative potential capacity expansion.

Figure 5: Specification B Total Transformation Elasticity Trajectory (θ_t^{SpecB}, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t^{\text{SpecB}} < 1.0$).



6 Summary of Key Empirical Findings & Takeaways

The empirical estimations establish four key findings for the dissertation committee:

1. **Distributionally Conditioned Mechanization Choice**: Under Initial-State centering with step controls, the interaction parameter $\hat{\theta}_{\omega\kappa}$ remains positive and statistically significant ($\hat{\theta}_{\omega\kappa} = 15.40^{***}$). Rising labor share drives intensive machinery accumulation relative to plant structures.

2. **Cointegration Hygiene and Variance Reduction**: Step controls ($\mathbf{D}_{1933}, \mathbf{D}_{1947}, \mathbf{D}_{1974}, \mathbf{D}_{2008}$) eliminate non-stationary residual drift (C_{Shin} drops from 0.261 to 0.042) and reduce residual variance ($\hat{\sigma}_u$ drops from 0.200 to 0.112).

3. **Stagnation Gap Recovery**: Reconstructed capacity utilization under Specification B ($\hat{\mu}_t^{\text{SpecB}}$) recovers a 50 percentage point capacity utilization gap post-2000 relative to aggregate Specification A, identifying post-GFC industrial stagnation driven by machinery investment decline.

4. **Severe Capital Overaccumulation Regime ($\theta_t^{\mathbf{SpecB}} < 1.0$)**: The total aggregate transformation elasticity θ_t^{SpecB} averages 0.41 over 1929–2024 and turns negative ($\theta_t^{\text{SpecB}} < 0$, reaching -0.27) in the post-2000 era. This confirms that the U.S. economy operates within a severe structural overaccumulation regime ($\theta_t < 1.0$), where additional capital accumulation yields diminishing or negative potential capacity output growth.